

Assignment 5

Case 1: Detecting Fraudulent Transactions in Banking

Context:

A bank is analyzing transaction records to detect possible fraudulent activities. Suspicious transactions are often extreme values in terms of amount. Read CSV file and perform the following tasks:

Tasks:

1. **Min-Max Method (user defined method):**
 - Normalize the "Amount" column
2. **IQR Method:**
 - Compute Q1, Q3, and IQR for the "Amount" column.
 - Identify outliers and replace them with the mean value.
3. **Z-Score Method (user defined method):**
 - Compute Z-scores for the "Amount" column.
 - Detect outliers using a Z-score threshold (± 3).
4. **Comparison:**
 - Compare the number of outliers detected using both methods.
 - Discuss which method is better for fraud detection and why.

Case 2: House Price Analysis

Context:

A real estate company wants to analyze house prices and detect extreme values that may indicate incorrect listings or luxury properties. CSV dataset is provided. Read and perform the following tasks:

Tasks:

1. **Robust Method:**
 - Normalize the "Price" column
2. **IQR Method:**
 - Compute Q1, Q3, and IQR for the "Price" column.
 - Identify outliers and replace them with the median value.
3. **Z-Score Method (user defined method):**
 - Compute Z-scores for the "Price" column.
 - Detect outliers using a Z-score threshold (± 3).
4. **Analysis:**
 - Which method performs better for detecting anomalies in house prices?
 - How would handling outliers affect machine learning models predicting house prices?

Case 3: Employee Salary Analysis

Scenario: You are working as an HR analyst and need to analyze salaries based on various attributes.

Employee_ID	Experience (Years)	Department	Salary (\$)	Education Level
E001	2	IT	55,000	Bachelor's
E002	10	Finance	85,000	Master's
E003	6	HR	65,000	Bachelor's
E004	15	IT	120,000	PhD

Tasks:

- 1. **Robust Normalization:** Apply robust scaling on "Experience (Years)".
- 2. **Z-score Standardization:** Standardize the "Salary" column.
- 3. **Ordinal Encoding:** Encode "Education Level" (Bachelor's < Master's < PhD).
- 4. **One-Hot Encoding:** Encode "Department".
- 5. **Label Encoding:** Apply label encoding on "Education Level".